

Lab 12 (CS): Input/Output

Innopolis University
Course of Operating Systems

Week 12 – Lab

Peripheral device

Background:

- A peripheral device is controlled by writing and reading its registers. Often, a device has multiple registers that can be accessed at consecutive addresses either in the memory address space or in the I/O address space. Each device connected to the I/O bus has a set of I/O addresses, called I/O ports. I/O ports can be mapped to physical memory addresses so that the processor can communicate with the device through instructions that work directly with the memory.

Peripheral device

- Get serial ports from `/proc/ioproports` using `sudo` permission and save the output to `ex1.txt`.
- What are `dma1`, `pic1` and `timer1`? What do they represent?

Note: If you are getting zero-valued addresses when accessing the file, then use `sudo` permission.

Exercise 1 (1/4)

- Write a C program `ex1.c` that gets keyboard events directly from the keyboard device and prints them to stdout.

Note: Try exploring

`/dev/input/by-path/platform-i8042-serio-0-event-kbd`

Exercise 1 (2/4)

- You have to use the file `/dev/input/by-path/platform-i8042-serio-0-event-kbd` (or a similar keyboard event character device file) for capturing keyboard events. You should endlessly read from the file to get all events
- You should use the `input_event` structure from `linux/input.h`¹
- Only `PRESSED`, `REPEATED` and `RELEASED` events should be handled.
- Print the output events in format: **PRESSED 0x0023 (35)**
 - Where **PRESSED** - type of event, **0x0023** and **(35)** are hex and decimal representation of event code respectively.
- The program should be executed using `sudo` permission.
- The program should be terminated by pressing `E+X`.
- Print and save the output to `ex1.txt`
- Save the code in `ex1.c`

¹<https://www.kernel.org/doc/Documentation/input/input.txt>

Exercise 1 (3/4)

- Modify previous program to output shortcuts: $P+E \rightarrow$ “I passed the Exam!”, $C+A+P \rightarrow$ “Get some cappuccino!” and one custom shortcut of your choice.

Exercise 1 (4/4)

- The program should print on the specified shortcuts in addition to the output events (“PRESSED”, “RELEASED”, “REPEATED”).
- Create the shortcut of your choice with custom message. The number of keys should be at least 2 and no more than 6.
- Print the available shortcuts at start of the program.
- Print and append the output to `ex1.txt`.
- Submit `ex1.txt` and `ex1.c`.
- The source code in `ex1.c` should be well-documented and explained.

End of lab 12 (CS)